

```

import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<UniversityMember> members = new ArrayList<>();

        Professor p1 = new Professor(101, "Dr. Ziaur Rahman",
            180000, "ICT", "OOP Pattern", 15);
        AdministrativeOfficer a1 = new AdministrativeOfficer(201,
            "Mr. Kamal", 40000, "Administration Office");

        members.add(p1);
        members.add(a1);

        for(UniversityMember member: members) {
            member.displayBasicInfo();
            member.performDuty();
        }

        for(UniversityMember member: members) {
            if(member instanceof Professor) {
                Professor p = (Professor) member;
                System.out.println("Professor Information:");
                p.displayBasicInfo();
                p.showResearchProfile();
            }
        }
    }
}

```



```
public void showResearchProfile() {  
    System.out.println("Research Area: " + researchArea);  
    System.out.println("Publications: " + publications);  
}
```

@ override

```
public void performDuty() {  
    System.out.println(name + " conducts research and  
        teaches advanced courses.");  
}
```

```
public class AdministrativeOfficer extends Employee {
```

```
    private String office;
```

```
    AdministrativeOfficer(int memberId, String name,
```

```
        double salary, String office) {
```

```
        super(memberId, name, salary);
```

```
        this.office = office;
```

```
    }
```

```
    public void showOffice() {
```

```
        System.out.println("Office: " + office);  
    }
```

@ Override

```
public void performDuty() {
```

```
    System.out.println(name + " manage
```

```
        administrative activities");  
}
```

```
}
```



```
public class AcademicStaff extends Employee {
```

```
    protected String department;
```

```
    public AcademicStaff(int memberId, String name,  
double salary, String department)
```

```
    super(memberId, name, salary);
```

```
    this.department = department;
```

```
}
```

```
    public void showDepartment() {
```

```
        System.out.println("Department: " + department);
```

```
    }  
    @Override
```

```
    public void performDuty() {
```

```
        System.out.println(name + " teaches students in the "
```

```
department + " department.");
```

```
}
```

```
public class Professors extends AcademicStaff {
```

```
    private String researchArea;
```

```
    private int publications;
```

```
    Professors(int memberId, String name, double salary,
```

```
String department, String researchArea, int  
publications) {
```

```
        super(memberId, name, salary, department);
```

```
        this.researchArea = researchArea;
```

```
        this.publications = publications;
```



```

abstract public class UniversityMember {
    protected int memberId;
    protected String name;
    public UniversityMember(int memberId, String name) {
        this.memberId = memberId;
        this.name = name;
    }
    public void displayBasicInfo() {
        System.out.println("ID: " + memberId);
        System.out.println("Name: " + name);
    }
    public abstract void performDuty();
}

```

```

public class Employee extends UniversityMember {
    protected double salary;
    public Employee(int memberId, String name, double salary) {
        super(memberId, name);
        this.salary = salary;
    }
    @Override
    public void performDuty() {
        System.out.println(name + " performs general duties");
    }
}

```